

Hands-on lab on Hadoop Cluster



What is a Hadoop Cluster?

A Hadoop cluster is a collection of computers, known as nodes, that are networked together to perform parallel computations on big data sets. The Name node is the master node of the Hadoop Distributed File System (HDFS). It maintains the meta data of the files in the RAM for quick access. An actual Hadoop Cluster setup involves extensive resources which are not within the scope of this lab. In this lab, you will use dockerized hadoop to create a Hadoop Cluster which will have:

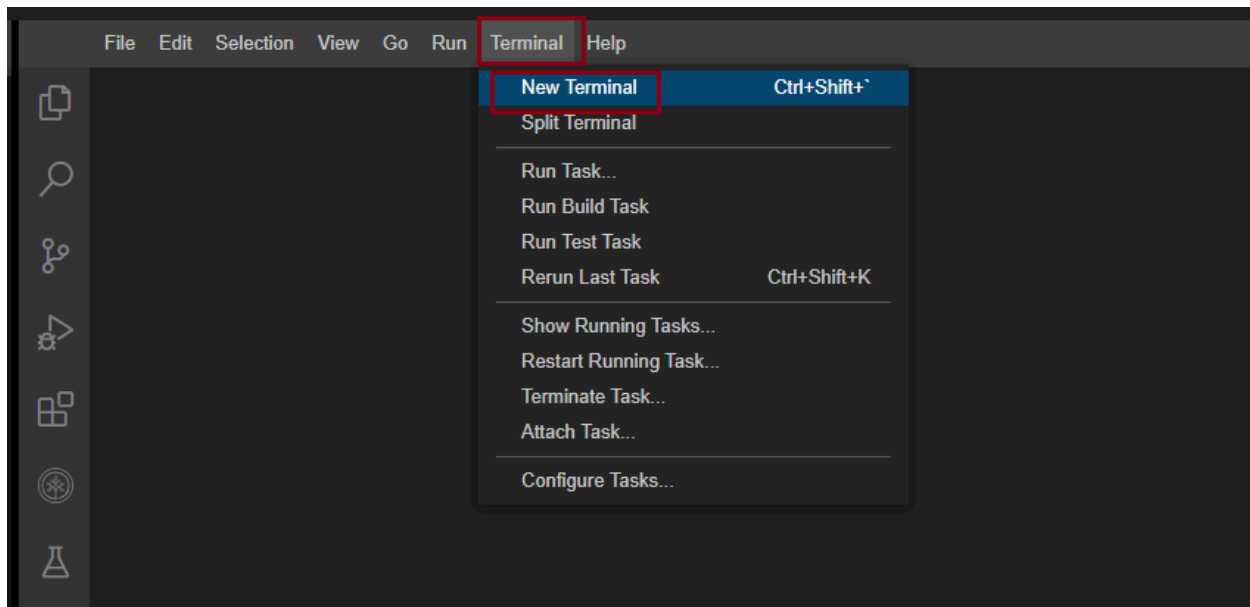
1. Namenode
2. Datanode
3. Node Manager
4. Resource manager
5. Hadoop history server

Objectives

- Run a dockerized Cluster Hadoop instance
- Create a file in the HDFS and view it on the GUI

Set up Cluster Nodes Dockerized Hadoop

1. Start a new terminal



1. Clone the repository to your theia environment.

```
git clone https://github.com/ibm-developer-skills-network/ooxwv-docker_hadoop.git
```

1. Navigate to the docker-hadoop directory to build it.

```
cd ooxwv-docker_hadoop
```

1. Compose the docker application.

```
docker-compose up -d
```

Compose is a tool for defining and running multi-container Docker applications. It uses the YAML file to configure the services and enables us to create and start all the services from just one configuration file.

You will see that all the five containers are created and started.



1. Run the namenode as a mounted drive on bash.

```
docker exec -it namenode /bin/bash
```

1. You will observe that the prompt changes as shown below.

```
theia@theiadocker-lavanyas:/home/project/docker-hadoop$ docker exec -it namenode /bin/bash
root@d72225e7724e:/#
```

Explore the hadoop environment

As you have learnt in the videos and reading thus far in the course, a Hadoop environment is configured by editing a set of configuration files:

- **hadoop-env.sh** Serves as a master file to configure YARN, HDFS, MapReduce, and Hadoop-related project settings.
- **core-site.xml** Defines HDFS and Hadoop core properties
- **hdfs-site.xml** Governs the location for storing node metadata, fsimage file and log file.
- **mapred-site.xml** Lists the parameters for MapReduce configuration.
- **yarn-site.xml** Defines settings relevant to YARN. It contains configurations for the Node Manager, Resource Manager, Containers, and Application Master.

For the docker image, these xml files have been configured already. You can see these in the directory `/opt/hadoop-3.2.1/etc/hadoop/` by running

```
1 ls /opt/hadoop-3.2.1/etc/hadoop/*.xml
```

Create a file in the HDFS

1. In the HDFS, create a directory structure named `user/root/input`.

```
1 hdfs dfs -mkdir -p /user/root/input
```

2. Copy all the hadoop configuration xml files into the input directory.

```
1 hdfs dfs -put $HADOOP_HOME/etc/hadoop/*.xml /user/root/input
```

3. Create a `data.txt` file in the current directory.

```
1 curl https://raw.githubusercontent.com/ibm-developer-skills-network/ooxwv-docker_hadoop/master/SampleMapReduce.txt
```

4. Copy the `data.txt` file into `/user/root/`.

```
1 hdfs dfs -put data.txt /user/root/
```

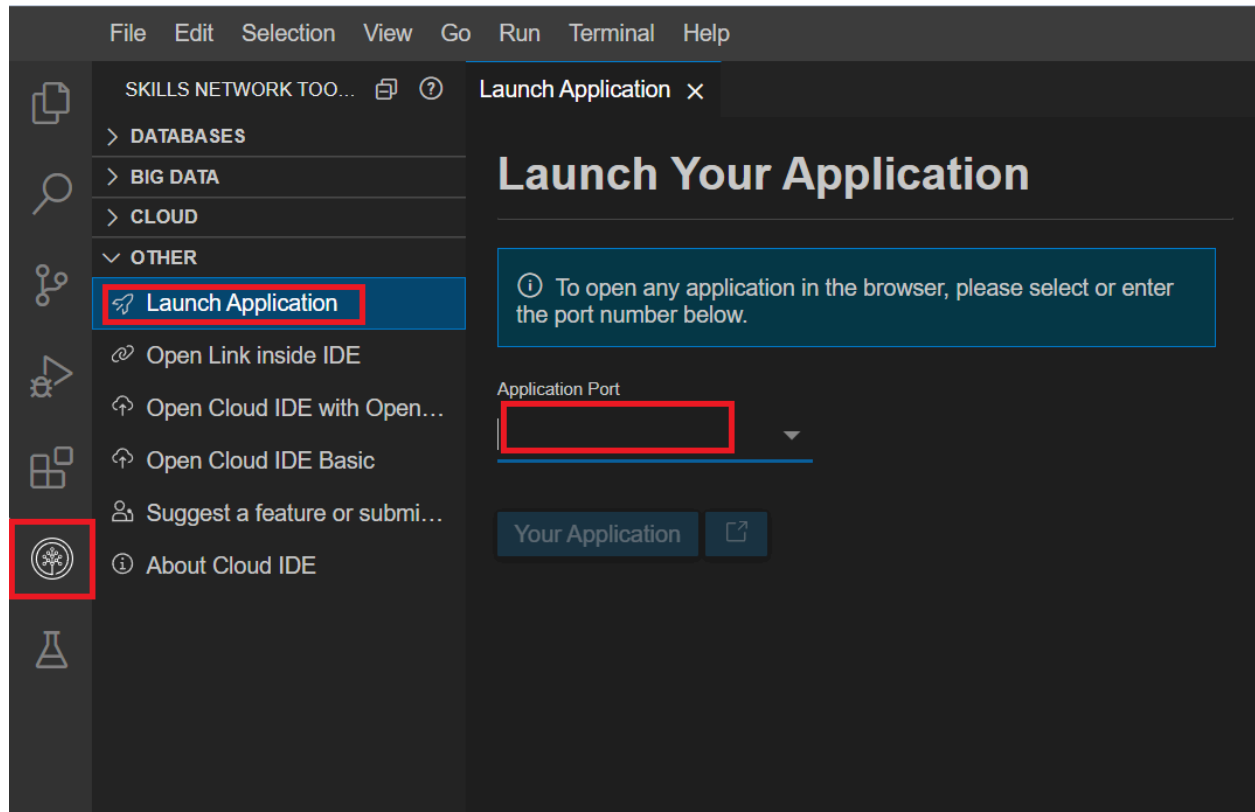
5. Check if the file has been copied into the HDFS by viewing its content.

```
1 hdfs dfs -cat /user/root/data.txt
```

View the HDFS

1. Click the button below or click on the Skills Network button on the left, it will open the "Skills Network Toolbox". Then click the Other then Launch Application. From there you should be able to enter the port number as **9870** and launch.

View HDFS



1. This will open up the Graphical User Interface (GUI) of the Hadoop node. Click on **Utilities** > **Browse the file system** to browse the files.

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Browse the file system

Logs

Log Level

Metrics

Configuration

Process Thread Dump

Overview 'namenode:9000' (active)

Started:	Mon Jul 12 15:11:20 +0530 2021
Version:	3.2.1, rb3cbbb467e22ea829b3808f4b7b01d07e0bf3842
Compiled:	Tue Sep 10 21:26:00 +0530 2019 by rohithsharmaks from branch-3.2.1
Cluster ID:	CID-0dba2137-1551-44b7-8ab3-49a6661cdaf7
Block Pool ID:	BP-936334794-172.18.0.2-1626082572639

1. View the files in the directories that you have just created by clicking on **user** then **root**.

Browse Directory

/

Go!

Show

25

 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	root	supergroup	0 B	Nov 10 16:00	0	0 B	rmstate	
☐	drwxr-xr-x	root	supergroup	0 B	Nov 10 16:01	0	0 B	user	

Showing 1 to 2 of 2 entries

Previous

1

Next

Hadoop, 2019.

Browse Directory

/user

Go!

Show

25

 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	root	supergroup	0 B	Nov 10 16:01	0	0 B	root	

Showing 1 to 1 of 1 entries

Previous

1

Next

Browse Directory

Show 25 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	-rw-r--r--	root	supergroup	6.7 KB	Jan 18 16:18	3	128 MB	data.txt	
☐	drwxr-xr-x	root	supergroup	0 B	Jan 18 16:16	0	0 B	input	

Showing 1 to 2 of 2 entries

1. Notice that the block size is 128 MB though the file size is actually much smaller. This is because the default block size used by HDFS is 128 MB.
2. You can click on the file to check the file into. It gives you information about the file in terms of number of bytes, block id etc.,

File information - data.txt



[Download](#)

[Head the file \(first 32K\)](#)

[Tail the file \(last 32K\)](#)

Block information --

Block 0 ▼

Block ID: 1073741839

Block Pool ID: BP-1800570971-172.18.0.5-1642502538329

Generation Stamp: 1015

Size: 6858

Availability:

- 1bb0a610767b

Close

Congratulations! You have:

- Deployed Hadoop using Docker
- Created data in HDFS and viewed it on the GUI



Tweet and share your achievement!